ACADEMIC CLINICAL DECISION-SUPPORT PROTOTYPE

MediAI

Smart Patient Triage & Emergency Risk Prediction System

Comprehensive Technical Architecture, System Engineering & Implementation Plan for Evaluation & Code Review

PRIMARY AUTHOR & DEVELOPER
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PROJECT REPOSITORY
https://github.com/jesica5/MediAI-Healthcare.git

TARGET ARCHITECTURE
Spring Boot 3.3.4 • MySQL 8.0 • Java 17+ • Vanilla HTML/CSS/JS

EVALUATION PHASE
Capstone Project Final Review & System Demonstration

CORE CLINICAL ENGINE
Deterministic Triage Engine + Google Gemini AI (2.5-Flash) + TreeSHAP

VERIFICATION STATUS
18/18 Automated JUnit Tests Passing • Build SUCCESS

Academic Medical Safety Disclaimer: MediAI is an educational clinical decision-support research system designed to evaluate multi-parameter physiological triage and explainability drivers. It must never independently diagnose patients, prescribe pharmaceuticals, compute drug dosages, or substitute qualified emergency physicians.

1. Executive Summary & Architecture Overview

MediAI is an end-to-end, full-stack hospital clinical triage and patient prioritization system engineered according to modern hospital emergency department protocols (Emergency Severity Index / Canadian Triage Acuity Scale). It bridges high-throughput front-end clinical workflows with persistent relational storage and multi-domain AI explainability.

1.1 Approved Technology Stack Alignment

The system strictly adheres to the approved capstone toolset requirements, ensuring full compliance without disallowed dependencies:

Tier	Technology	Version / Specification	Role in System
Front-End	HTML5, Vanilla CSS3, Modern JavaScript	ECMAScript 2022 / CSS Variables	Accessible, high-performance UI matching clinical terminal guidelines without third-party design bloat.
Templates	Thymeleaf & Static Resource Engines	Spring Boot 3.3.4 Starter	Dual-serving of clinical views (intake terminal, queue, doctor workstation, analytics).
Back-End	Java Core, Spring Boot RESTful Web Services	Java 17 (LTS), Spring Boot 3.3.4	Enterprise service architecture, dependency injection, and deterministic clinical engines.
ORM & Persistence	Spring Data JPA & Hibernate Core	Hibernate 6.5.3, JPA 3.1	Entity-relational mapping, transactional data consistency, derived queries, and relationship integrity.
Database	MySQL Relational Database Server	MySQL 8.0+ / InnoDB Engine	Persistent, ACID-compliant transactional storage for patients, vitals, triage records, and audit journals.
Build & Lifecycle	Apache Maven	Maven 3.9+ via `mvnw` wrapper	Dependency resolution, multi-stage compilation, and automated test execution.

1.2 Core Architectural Principles & Strict UI Preservation

> **Zero UI Disruption Mandate:** All new capabilities (Voice Input, Notifications, 2-Step OTP, Feedback Ratings, and AI Clinical Chatbot) are added in a 100% backward-compatible manner. Existing layouts, colors, fonts, routes, and triage risk algorithms remain completely intact.

- **Deterministic Safety First:** Machine learning inferences are strictly bounded by hard deterministic physiological tripwires (e.g., SpO2 < 90% immediately overrides all models to P1 Critical).
- **Zero Protected Health Information (PHI) Leakage:** Patient names are encrypted at rest with AES-256-GCM. Medical Record Numbers (MRNs) use SHA-256 salted hashes. System audit journals log zero raw patient health identifiers.
- **Bilingual Clinical Accessibility:** Full dynamic language interchange between English (`en-US`) and Tamil (`ta-IN`) across user interfaces, voice recognition, and clinical reassurance output.

1.3 System Component Architecture

Clinical Decision Support Layer

• **TriageEngineService:** Computes scaled risk scores (0-100) and priority categories (P1 to P4).
• **GeminiAiService:** Evaluates prompt-guarded contextual clinical rationale.
• **TreeSHAP Factor Breakdown:** Transparent attribution weights explaining why an acuity tier was assigned.

Security & Audit Layer

• **JwtAuthFilter & SecurityConfig:** Stateless bearer token management.
• **AuthService & OTP Engine:** SHA-256 hashed 2-factor passcodes with 5-min timeout and attempt limiting.
• **AuditService:** Immutable journal logging actor roles, IP addresses, and encrypted action categories.

2. Review of Core Features: Feature 1 & Feature 2

2.1 Feature 1: Voice Input for Clinical Symptoms

The voice recognition module enables patients and triage triage nurses to dictate acute symptoms hands-free, which is especially vital in emergency triage where manual keyboard entry can cause critical intake delays.

> **Implementation Location:** Integrated seamlessly adjacent to the chief complaint input field in `intake.html` (line 230) and `profile.html` (line 112) via a compact microphone button (`btn btn-outline btn-sm`).

Requirement	Implementation Mechanism	Verification Status
Bilingual Speech Recognition	Leverages Web Speech API (`MediAPI.voice.createRecognizer`) configured to stream either `en-US` or `ta-IN` dynamically based on active language selection.	VERIFIED
Editable Confirmation	Transcribed speech streams into the editable `ptComplaint` text input. Patients and nurses can review, adjust, or delete text before advancing. Speech is never submitted automatically.	VERIFIED
Manual Typing Fallback	Input field retains standard keyboard focus. Manual typing remains 100% available without interruption.	VERIFIED
Permission & Error Safety	Catches `not-allowed` , `no-speech` , and unsupported browser states with polite inline non-blocking guidance banners.	VERIFIED
Preservation of Triage Logic	Captured text flows into the existing `POST /api/v1/triage/assess` pipeline without altering clinical calculations.	VERIFIED

2.2 Feature 2: Real-Time Clinical Notifications & Alerts System

A persistent, role-aware notification subsystem provides automated clinical alerts for intake completion, triage tier status updates, resuscitation red flags, and doctor override events.

Database & Entity Structure

- **Entity:** `Notification.java` mapping to table `notifications` .
- **Fields:** `id` , `targetRole` (ALL, DOCTOR, NURSE), `title` , `message` , `priority` (CRITICAL, HIGH, INFO), `isRead` , `referenceId` , `createdAt` .
- **Repository:** `NotificationRepository.java` supporting unread counting and role filtering.

UI Top-Nav Bell Drawer

- **Global Injector:** `MediAPI.initNotifications()` auto-mounts a bell (🔔) in `.nav-actions` on every page.
- **Badge:** Real-time unread counter pill.
- **Drawer:** Interactive dropdown showing timestamps, priority color indicators, individual "Mark read" triggers, and "Mark all read" button.

Automated Alert Triggers Workflow

Clinical Trigger Event	Target Role	Priority Tier	Notification Content
Patient Intake Assessment Completed	ALL	INFO	"New Triage Assessment Recorded with [Category] priority."
Tier-1 Hard Red Flag / Critical Hypoxia Triggered	DOCTOR / NURSE	CRITICAL	"🚨 CRITICAL Triage Alert: Immediate resuscitation bay attention required."
Physician Overrides AI Triage Classification	ALL	HIGH	"👩‍⚕️ Physician Triage Override: Category updated to [Target Category]."

3. Review of Core Features: Feature 3 & Feature 4

3.1 Feature 3: Login Authentication, Role Authorization & 2-Step OTP Security

Staff authentication portal (`signin.html`) provides enterprise-grade Two-Step Verification (2FA) with cryptographically secure One-Time Passcodes (OTP) while completely preserving the existing visual layout.

Two-Step Verification Protocol

1. **Step 1:** User submits email and password. Server validates credentials against BCrypt hashes in MySQL.
2. **Step 2:** Generates a 6-digit random passcode, cryptographically hashes it with SHA-256, and stores it in `user_otps` with a 5-minute expiry timestamp.
3. **Client Challenge:** Form transitions in-place to Step 2 showing masked email (`d***r@mediai.org`), countdown timer, and OTP input field.
4. **Validation:** Submitting OTP verifies hash match. Upon success, OTP is immediately invalidated and JWT token is issued.

Security Guardrails & Hardening

- **Zero Cleartext Storage:** Passcodes are hashed with SHA-256 before database insertion.
- **Attempt Limiting:** Maximum 3 failed verification attempts before the session token is permanently revoked.
- **Single-Use Enforcement:** Re-using a validated OTP returns HTTP 400 Bad Request.
- **Resend Rate Limiting:** 60-second cooldown period prevents SMS/email flooding.
- **Backend RBAC:** Endpoints enforce role verification via JWT claims and request headers.

3.2 Feature 4: Patient Care Experience & Feedback Ratings Registry

An integrated clinical quality rating and feedback subsystem allows patients and clinicians to submit verified care evaluations following triage intake without modifying medical calculations.

> **Clinical Fire-walling:** Feedback scores and qualitative comments are strictly segregated from AI risk scores, TreeSHAP algorithms, and triage classifications. They are used purely for clinical operations and service quality telemetry.

Component Architecture

Subsystem	Implementation Details	Location
Feedback Form	Interactive 1-to-5 star rating selector + sanitized text comment area (capped at 1,000 characters to prevent buffer abuse). Appears immediately after triage calculation.	Step 4 in `intake.html`
Persistent Entity	`Feedback.java` stored in MySQL `feedbacks` table with `rating` , `comment` , `submitterRole` , `category` , `createdAt` .	`com.mediai.entity.Feedback`
Admin Telemetry Registry	Displays aggregate metrics: average star rating, total submission counter, star distribution percentage bars (5★ through 1★), and real-time qualitative comment list.	Card in `analytics.html`
REST Endpoints	`POST /api/v1/feedback` (submission) and `GET /api/v1/feedback/summary` (aggregation).	`FeedbackController.java`

4. Review of Feature 5 & Complete Database Schema

4.1 Feature 5: Clinical AI Chatbot Assistant (MediAI Assistant)

A universal, non-intrusive floating clinical widget (`#mediaiChatbotLauncher`) mounted across all pages to guide patients and clinicians with symptom inquiries, normal vital sign baselines, and emergency triage navigation.

Clinical Intelligence & Safety Interception

• **Red-Flag Emergency Interception:** Automatic real-time regex detection of life-threatening presentations (crushing chest pain, severe dyspnea, heavy hemorrhage, stroke symptoms) in English & Tamil.
• **Emergency Directive:** Immediately flags `is_emergency = true` , halts standard Q&A, and issues prominent directive to dial **108 / 112 / 911** or proceed directly to the Emergency Room.
• **AI Generative + Clinical Fallback:** Seamless fallback covering fever thresholds (≥ 38.5°C), clinical pain scales (1-10), and vital signs baselines.

Interactive Widget Capabilities

• **Bilingual Interchange:** Instant toggle between English (`EN`) and Tamil (`தமிழ்`) updating prompt context and chips.
• **Integrated Voice Dictation:** Built-in microphone button powered by Web Speech API allowing hands-free patient voice queries.
• **Quick Action Chips:** One-click prompts for common medical inquiries.
• **Session Memory:** Conversations persistently stored in MySQL `chat_messages` with history retrieval and clearing APIs.

4.2 Relational Database Schema Architecture (MySQL 8.0)

All data entities are managed using Hibernate with `spring.jpa.hibernate.ddl-auto=update` , guaranteeing zero data loss on restart and maintaining relational integrity:

Table Name	Primary Key	Key Columns & Constraints	Relational Purpose
`patients`	`id` (VARCHAR 36)	`mrn_hash` , `encrypted_name` , `dob` , `gender`	Master patient demographics with AES-256 encrypted names.
`vital_signs`	`id` (VARCHAR 36)	`patient_id` (FK), `heart_rate` , `systolic_bp` , `spo2` , `temp`	Physiological vital readings with validation bounds.
`triage_assessments`	`id` (VARCHAR 36)	`patient_id` (FK), `vitals_id` (FK), `calculated_category` , `risk_score`	Acuity classification with TreeSHAP and AI rationale payloads.
`triage_reviews`	`id` (VARCHAR 36)	`assessment_id` (FK), `doctor_id` , `final_category` , `is_overridden`	Doctor overrides and clinical confirmation audits.
`notifications`	`id` (VARCHAR 36)	`target_role` , `title` , `message` , `priority` , `is_read`	Role-based persistent clinical alerts and triage updates.
`user_otps`	`id` (VARCHAR 36)	`temp_token` (IDX), `otp_hash` , `attempts` , `expires_at`	Single-use SHA-256 hashed 2FA passcode sessions.
`feedbacks`	`id` (VARCHAR 36)	`rating` (1-5), `comment` , `submitter_role` , `reference_id`	Patient care experience and staff feedback records.
`chat_messages`	`id` (VARCHAR 36)	`conversation_id` (IDX), `sender` , `message` (TEXT), `is_emergency`	Bilingual clinical assistant conversation history.
`audit_logs`	`id` (VARCHAR 36)	`action_category` , `actor_id` , `ip_address` , `timestamp`	Zero-PHI append-only compliance journal trail.

5. RESTful API Catalog & Verification Results

5.1 Comprehensive RESTful Service Endpoints

Method	Endpoint Route	Access / Role	Description & Purpose
`POST`	`/api/v1/auth/login-step1`	Public	Validates credentials; issues temp token & dispatches 6-digit OTP.
`POST`	`/api/v1/auth/verify-otp`	Public	Verifies OTP hash; enforces attempt limits; issues JWT session token.
`POST`	`/api/v1/auth/resend-otp`	Public	Issues replacement OTP with cooldown rate-limit verification.
`POST`	`/api/v1/triage/assess`	Public / Intake	Computes triage category, risk score, TreeSHAP & dispatches alert.
`GET`	`/api/v1/notifications`	Authenticated	Retrieves notifications filtered by role (`?role=ALL/DOCTOR/NURSE`).
`PUT`	`/api/v1/notifications/read-all`	Authenticated	Marks all unread notifications as read.
`POST`	`/api/v1/feedback`	Public / User	Submits 1-5 star care rating and sanitized qualitative comment.
`GET`	`/api/v1/feedback/summary`	Admin / Staff	Returns aggregate average, distribution bars, and recent comments.
`POST`	`/api/v1/chat/message`	Public / Patient	Interprets clinical query, intercepts red flags, returns AI reply.
`POST`	`/api/v1/doctor/override`	Doctor / Staff	Physician confirmation/override with 15-char rationale safety check.

5.2 Verification Suite & Test Results Summary

> **Automated Test Status:** All 18 automated JUnit 5 unit and integration tests passed cleanly. Complete integration scripts (`verify_features.ps1` and `verify_chatbot.ps1`) executed with zero errors.

Test Suite	Tests Run	Pass Rate	Key Validations Tested
`AgeCalculationTest`	4	100%	Leap year math, future date rejection, dynamic age badge updates.
`VitalSignsValidationTest`	5	100%	Physiological bounds (HR 20-300, BP, SpO2, Temp), Diastolic < Systolic check.
`TriageEngineTest`	4	100%	Hard red-flag tripwires, scaled risk score math, TreeSHAP impact drivers.
`GeminiAiServiceTest`	3	100%	Prompt-guarded payload parsing, deterministic clinical fallback.
`DoctorServiceTest`	2	100%	15-character rationale enforcement on Critical downgrade, notification trigger.
`verify_features.ps1`	6 Flow Stages	100%	E2E check: HTTP delivery, 2FA OTP, Notifications, Triage, Feedback, Doctor review.
`verify_chatbot.ps1`	6 Flow Stages	100%	Bilingual greeting, symptom Q&A, Tamil queries, Emergency 108 interception, DB history.

5.3 Demonstration & Reviewer Run Instructions

Launch Commands

1. Verify MySQL is running on `localhost:3306`.
2. Execute Spring Boot application:
`.\mvnw.cmd spring-boot:run`
3. Open browser: `http://localhost:8080`

Pre-configured Credentials (1-Click Fill)

- **Doctor Workstation:** `doctor@mediai.org` / `Doctor@123`
- **Triage Nurse:** `nurse@mediai.org` / `Nurse@123`
- **Administrator:** `admin@mediai.org` / `Admin@123`